ASSIGNMENT – 2

(1) Velocity of 1st particle wrt ground = v (x-dirn),
" " 2nd " " = v (y-dirn).

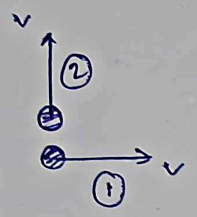
Applying S.L.T. w.r.t particle 1:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \; ; \quad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}} \; ; \quad y = y' ; \quad z = z'$$

Velocity expression in 1st frame;

$\boxed{x\text{-dirn}}$:

$$\frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt} = \left[\frac{\frac{dx}{dt} - v}{\cancel{\sqrt{1 - v^2/c^2}}}\right] \cdot \frac{\cancel{\sqrt{1 - v^2/c^2}}}{1 - \frac{v}{c^2}\left(\frac{dx}{dt}\right)}$$



$$\Rightarrow v_x' = \frac{\left(\frac{dx}{dt} - v\right)}{\left(1 - \frac{v}{c^2}\frac{dx}{dt}\right)}$$

$$= \frac{0 - v}{1 - \frac{v}{c^2}(0)} = \boxed{-v = v_x'}$$

$\boxed{y\text{-dirn}}$:

$$\frac{dy'}{dt'} = v_y' = \frac{dy'/dt}{dt'/dt}$$

$$= \frac{dy'/dt}{1 - \frac{v}{c^2}\frac{dx}{dt}} \cdot \sqrt{1 - \frac{v^2}{c^2}}$$

$$= \frac{v}{1 - \frac{v^2}{c^2}(0)} \cdot \sqrt{1 - \frac{v^2}{c^2}} \qquad \cancel{= \frac{v}{\sqrt{1 - \frac{v^2}{c^2}}}}$$

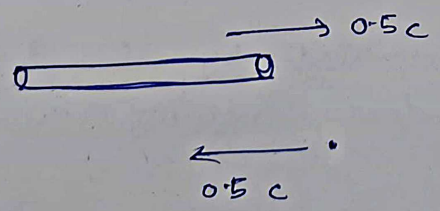
$$\boxed{v_y' = v\sqrt{1 - \frac{v^2}{c^2}}}$$

∴ Final velocity of 2 wrt 1 :

$$v' = \sqrt{v_x'^2 + v_y'^2}$$

$$= \sqrt{v^2 + v^2\left(1 - \frac{v^2}{c^2}\right)} = v\sqrt{2 - \frac{v^2}{c^2}} \quad \text{(Ans)}$$

(2)



Ground Frame:

v_p = velocity of particle
v_r = velocity of rod.

Let the particle take t time to cross the rod.

$$\therefore \quad v_p t = L - v_r t$$

$$\Rightarrow \boxed{t = \frac{L}{v_p + v_r}}$$

$L = \frac{L_0}{\gamma}$ [L_0 = proper length].

$= \cancel{0.1\sqrt{1-\frac{v_r^2}{c^2}}}$ $\quad 0.1\sqrt{1-\frac{v_r^2}{c^2}}$

$= 0.1\sqrt{1-0.25} = 0.1\sqrt{0.75} =$

$= 0.1 \times 0.866$

$= 0.0866$ m

$\therefore t = \frac{0.0866}{0.5c + 0.5c} = \frac{0.0866}{c}$ s (Ans)

Rod's Rest Frame:

~~x' = position of particle~~ ; ~~t' = time in rod's~~

~~x = position of rod~~

$$\frac{dx'}{dt'} = \frac{\frac{dx}{dt} - v}{1 - \frac{v}{c^2}\frac{dx}{dt}}$$

$$= \frac{-0.5c - 0.5c}{1 - \frac{0.5c}{c^2}(-0.5c)} = \frac{-c}{1+0.25} = -0.8c$$

Time taken by particle in rod's rest frame

$= \frac{L_0}{-0.8c} = \frac{10\times10^{-2}}{0.8c} = \frac{0.125}{c}$ s (Ans)

Particle's Rest Frame:

Rod's velocity in particle's rest frame = 0.8 c

Length of rod in particle's frame $= L_0\sqrt{1-\frac{v^2}{c^2}}$

$= 10\sqrt{1-(0.8)^2} \times 10^{-2}$

$= 6\times10^{-2}$ m

Time taken $= \frac{6\times10^{-2}}{0.8c} = \frac{3}{4}\times10^{-1} = \frac{0.075}{c}$ s

(3) 2 frames S & S' are connected by S.L.T. if they are moving with constant velocity v wrt each other.

$x' = \frac{x - vt}{\sqrt{1-v^2/c^2}}$; $t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-v^2/c^2}}$; $y = y'$; $z = z'$

$$\frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt}$$

$$(v_x') = \frac{\frac{dx}{dt} - v}{\cancel{\sqrt{1-\frac{v^2}{c^2}}}} \cdot \frac{\cancel{\sqrt{1-v^2/c^2}}}{1-\frac{v}{c^2}\frac{dx}{dt}}$$

$$\frac{dx'}{dt'}(v_x') = \frac{\frac{dx}{dt} - v}{1-\frac{v}{c^2}\frac{dx}{dt}}$$

$$\frac{dy'}{dt'}(v_y') = \frac{dy/dt}{1-\frac{v}{c^2}\frac{dx}{dt}} \quad ; \quad \frac{dz'}{dt} = \frac{dz/dt}{1-\frac{v}{c^2}\frac{dx}{dt}} \quad (v_z')$$

$$\frac{d^2x'}{dt'^2} = \frac{d}{dt'}\left(\frac{dx'}{dt'}\right)$$

$$= \frac{d\left(\frac{dx'}{dt'}\right)/dt}{dt'/dt}$$

$$= \frac{d}{dt}\left[\frac{\left(\frac{dx}{dt} - v\right)}{1-\frac{v}{c^2}\frac{dx}{dt}}\right] \cdot \frac{\sqrt{1-\frac{v^2}{c^2}}}{1-\frac{v}{c^2}\frac{dx}{dt}}$$

$$= \frac{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)\left[\frac{d^2x}{dt^2}\right] - \left(\frac{dx}{dt} - v\right)\left(-\frac{v}{c^2}\frac{d^2x}{dt^2}\right)}{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)^{3/2}} \cdot \sqrt{1-\frac{v^2}{c^2}}$$

$$= \frac{\frac{d^2x}{dt^2}\left[1 - \cancel{\frac{v}{c^2}\frac{dx}{dt}} + \cancel{\frac{v}{c^2}\frac{dx}{dt}} - \frac{v^2}{c^2}\right]}{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)^{3/2}} \cdot \sqrt{1-\frac{v^2}{c^2}}$$

$$\boxed{a_x' = \frac{d^2x}{dt^2}\frac{\left(1-\frac{v^2}{c^2}\right)^{3/2}}{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)^{3/2}}}$$

$$\frac{d^2y'}{dt'^2} = \frac{d}{dt'}\left(\frac{dy'}{dt'}\right)$$

$$= \frac{d\left(\frac{dy'}{dt'}\right)/dt}{dt'/dt}$$

$$= \frac{d}{dt}\left[\frac{dy/dt}{\sqrt{1-\frac{v}{c^2}\frac{dx}{dt}}}\right] \cdot \frac{\sqrt{1-v^2/c^2}}{1-\frac{v}{c^2}\frac{dx}{dt}}$$

$$a_y' = \frac{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)\left(\frac{d^2y}{dt^2}\right) + \left(\frac{dy}{dt}\right)\left(\frac{v}{c^2}\frac{d^2x}{dt^2}\right)}{\left(1-\frac{v}{c^2}\frac{dx}{dt}\right)^{3/2}} \cdot \sqrt{1-\frac{v^2}{c^2}}$$

Similarly :

$$a_z' = \frac{\left(1 - \frac{v}{c^2}\frac{dx}{dt}\right)\left(\frac{d^2z}{dt^2}\right) + \left(\frac{dz}{dt}\right)\left(\frac{v}{c^2}\frac{d^2x}{dt^2}\right)\sqrt{1 - v^2/c^2}}{\left(1 - \frac{v}{c^2}\frac{dx}{dt}\right)^3}$$